

Activity 1.1.5 Top Gear

PLTW

TOP GEAR

≡ No Need for Speed

≡ Simple Gear Trains

≡ Practice!

≡ Rotation Direction

≡ Go for Slow

≡ Compound Gear Trains

≡ Conclusion

No Need for Speed

GOALS	MATERIALS	RESOURCES
• Design, construct, and solve for simple and compound gear trains. • Apply knowledge of gears and gear ratios to design gear trains to optimize speed or torque.		
GOALS	MATERIALS	RESOURCES
• Ruler • VEX® V5 POE Kit • Permanent marker		

GOALS	MATERIALS	RESOURCES
		• <u>VEX® Parts</u> • <u>VEX® V5 Build Support Guides</u> • <u>VEX® V5 Brain</u> • <u>VEX® Simple Motor Code</u> • <u>VEX® V5 Troubleshooting</u> • <u>Activity 1.1.5 Top Gear (Downloadable PDF)</u>

Fran’s farm is starting to gain some traction. She just purchased this vintage tractor so that she can “turn up dirt” (turning over and breaking up the soil) on the farm. Have you ever seen a tractor like this in operation?

They tend to move quite slowly. The engine might spin at 700 rotations per minute (rpm), but the five-foot-diameter tires certainly don’t spin that fast. (If they did, this tractor would cruise at 125 miles per hour!) How, then, is the engine able to rotate at one speed, while the tires rotate at another speed? [**Gears!**](#)



In this activity, you will explore how gears are used to optimize either speed or [torque](#). You'll construct both simple drive trains, as well as more complicated assemblies. Ultimately, you will make a small vehicle. Your goal is to make the slowest car possible. Start your engines!

Simple Gear Trains

1

Find the following VEX® gears, as shown in Figure 1.

- 12-tooth
- 36-tooth (x2)
- 60-tooth
- 84-tooth



Figure 1. 12, 36, 60, and 84-tooth VEX gears

2

Color the side of one tooth on each gear, as shown in Figure 2.

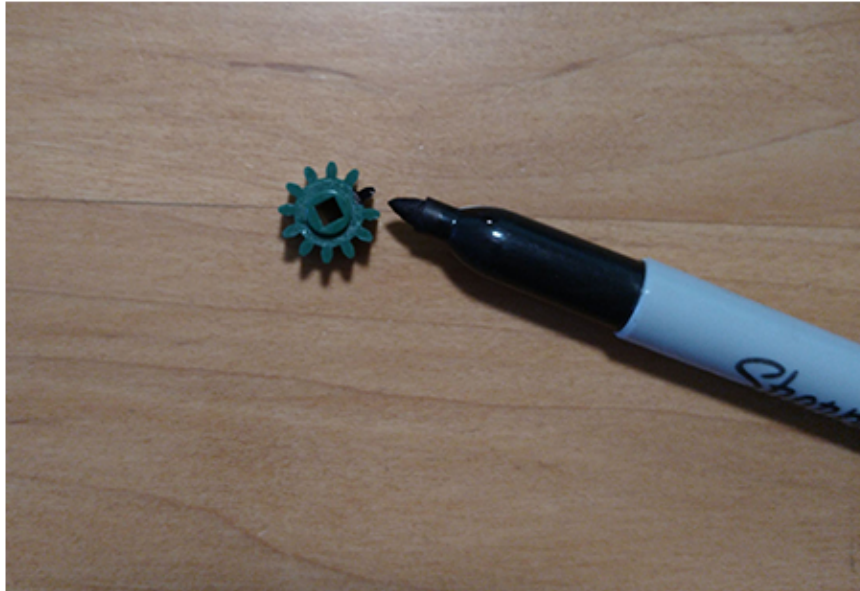


Figure 2. 12-tooth Gear with One Tooth Marked in Black

3

Mesh the two 36-tooth gears so that the marked teeth touch each other as shown in Figure 3.



Figure 3. Two 36-tooth VEX Gears Meshed, Marked Teeth Touching

Turn and Talk



If you turned the left gear one complete rotation, how many times would the gear on the right rotate? Explain your reasoning.

4

Test your prediction by turning the left gear one rotation.
Do the marked teeth line up again?

In steps 3 and 4, you modeled a **gear train**, which transfers energy from one axis to another. When gears are meshed in a row, each having its own axis of rotation, this is called a simple gear train. Since you supplied power to the left gear by turning it with your finger, this was the driver gear.

A **driver gear** typically has a motor or some other device connected to its axle to initiate rotation. The right gear was the **driven gear**. A simple gear train is shown in Figure 4.

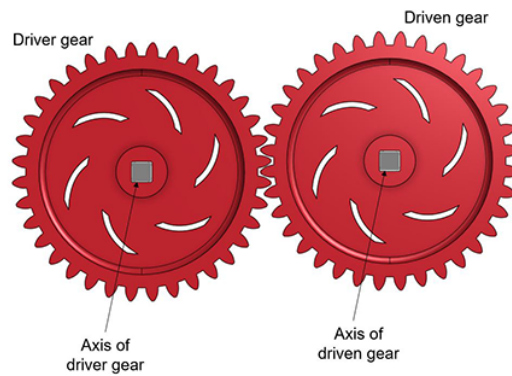


Figure 4. Simple Gear Train

5

Mesh a 36-tooth gear (left) with a 12-tooth gear (right) so that the marked teeth touch each other, as shown in Figure 5.



Figure 5. 36-tooth Gear Meshed with 12-tooth Gear

Turn and Talk



If you turned the gear on the left one complete rotation, how many times would the gear on the right rotate? Explain your reasoning.

Gear trains are often described by their **gear ratio**, which is a comparison between the driver and driven gears. You can calculate the gear ratio by dividing the number of teeth in the driven gear by the number of teeth in the driver gear. For this gear train, the gear ratio is 12/36 or 1/3. Gear ratios are often expressed using a colon, like a 1:3 (pronounced “one to three”) ratio.

$$\text{Gear Ratio} = \frac{\text{Teeth}_{\text{Driven}}}{\text{Teeth}_{\text{Driver}}}$$

Where:

Driver = Input Gear

Driven = Output Gear

6

Using a ruler, measure the diameter of each gear to the nearest half-inch.

7

Compare the gear ratio (as determined by the number of teeth of each gear) with the ratio of the gears' diameters.

Turn and Talk

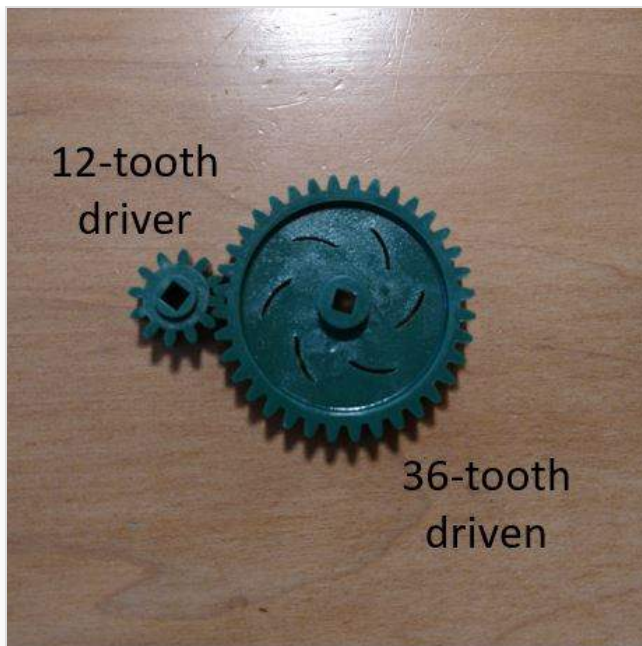


How does the gear ratio compare to the ratio of the diameters? Why do you think this is the case? How are gears and levers related?

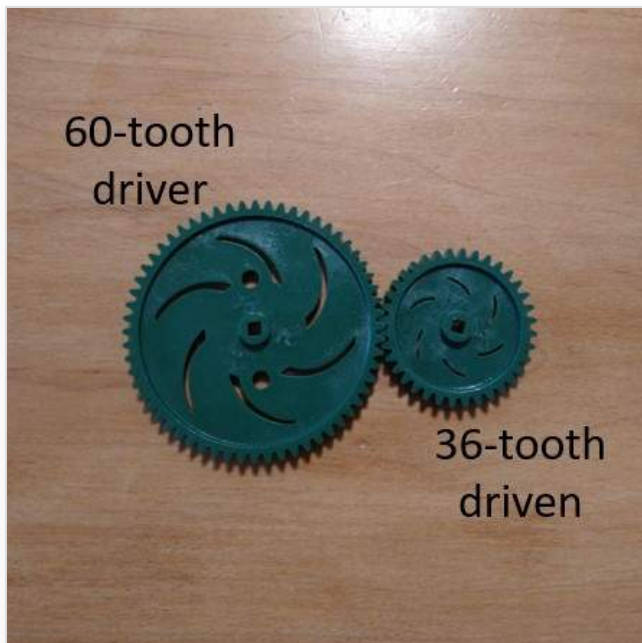
Practice!

8

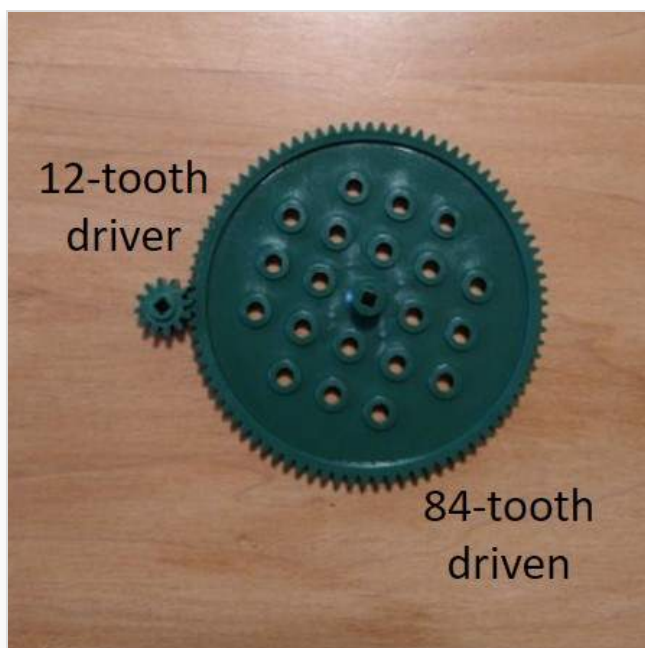
Make each of the following gear systems. Determine the gear ratio, and whether they are set up to optimize torque or speed. Rotate the gears to determine whether they behave the way you expect.



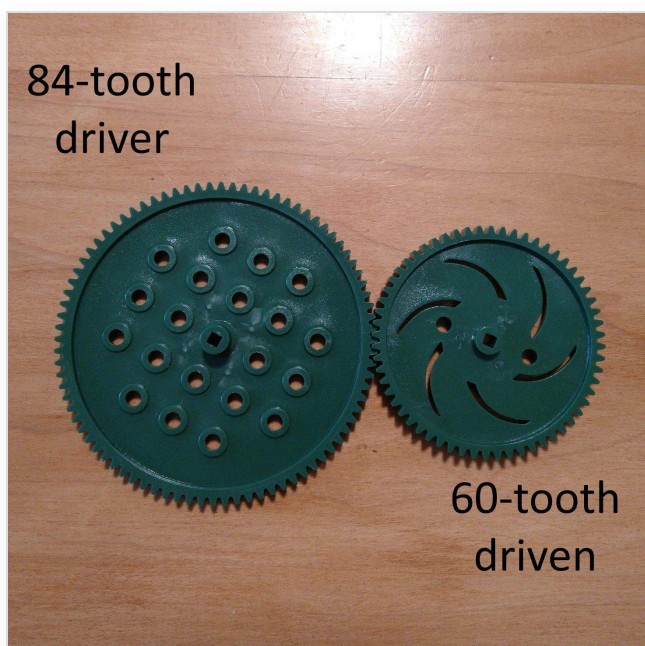
Gear Ratio: 3:1
Optimize: Torque



Gear Ratio: 36:60 Gear Ratio (or
0.6:1)
Optimize: Speed



Gear Ratio: 84:12 (or 7:1)
Optimize: Torque



Gear Ratio: 60:84 (or 0.7:1)
Optimize: Speed

Rotation Direction

In both of the simple gear trains you modeled, the driver and driven gears rotated in opposite directions. If you rotate the driver gear clockwise, the driven gear rotates counterclockwise.

9

With your partner, brainstorm a way to construct a simple gear train where the driver and driven gears rotate in the same direction. Use additional VEX components as necessary.

Need a Hint?

Try adding components to your gear train.

The gear that was added is called an [idler gear](#). While it changes rotation, how might it affect the gear ratio?

10

Arrange two 36-tooth gears as driver and driven gears with a 12-tooth gear as an idler gear, as shown in Figure 6.

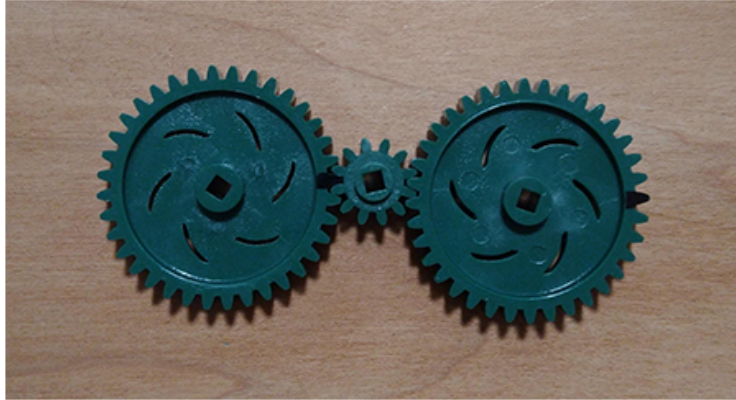


Figure 6. 36-tooth driver and driven gears with 12-tooth idler gear

Turn and Talk



Predict how many rotations of the driven gear will occur for each rotation of the driver gear. Explain your reasoning using mathematics.

11

To test your prediction, turn the left gear one rotation.

Go for Slow

12

Use what you know about simple gear trains to design and build the slowest tractor possible to race against your classmates. Use [VEX® Parts](#) and [VEX® V5 Build Support Guides](#) to help understand how best to connect VEX axles, bearings, and metal pieces.

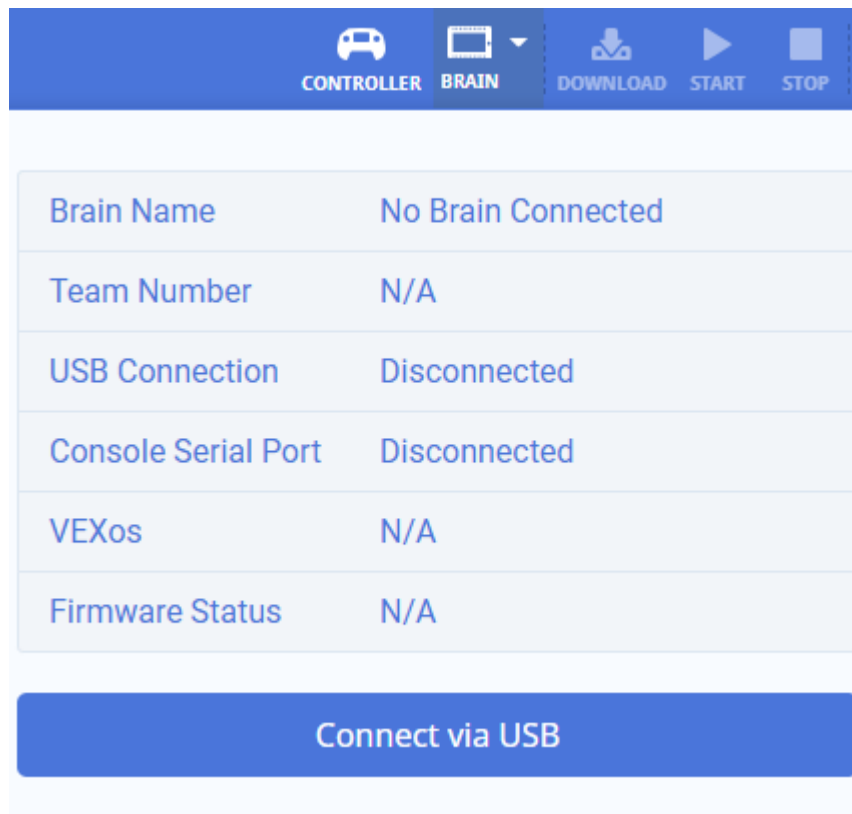
NOTE: This vehicle must include a [VEX® V5 Brain](#), a VEX V5 Battery Pack, a VEX V5 Smart Motor connected to Port 5 of the Brain, and a VEX V5 Limit Switch connected to Port A of the Brain.

13

Once you've constructed the vehicle, download the [VEX® Simple Motor Code](#) to your computer. Note the location. You will load this onto the VEX V5 Brain to control your vehicle.

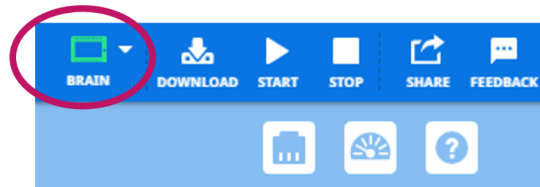
- Ensure the motor is connected to Port 5 and the switch is connected to Port A.
- Connect the USB cable from the computer to the Brain.
- Connect a VEX Battery Pack to the Brain.

- Navigate to **VEXcode V5**.
- In the top left of the window, select **File > Load From Your Device**.
- Navigate to the VEX Simple Motor Code that you downloaded at the beginning of step 13.
- Turn on the Brain.
- From the menu bar in the top right corner, select the **BRAIN** icon.
- On the box that opens, select **Connect**.



- On the box that opens, select **Continue**.
- On the next box, select **VEX Robotics Communications Port ...** and then **Connect**. The Brain

is successfully connected when the **BRAIN** icon in the code editor turns green.



- Select **DOWNLOAD**, adjacent to the BRAIN icon in the menu bar.

NOTE: Ensure your vehicle is in a safe state, meaning if the motor starts spinning, the vehicle won't get damaged. Consider raising a drive wheel off the surface, or disengaging any drivetrain gears by sliding them down an axle so they no longer mesh.

- To start your program, select **START** in the menu bar. You can turn the motor on and off using the limit switch. You can disconnect the USB cable from your Brain, and still run the program, now that it is downloaded. Consult [**VEX® V5 Troubleshooting**](#) if you are unable to run your program.

Compound Gear Trains

14

Using two 12-tooth gears and two 36-tooth gears, design a gear train that has a gear ratio of greater than 3:1. Since you are using so many gears, it will be difficult to rotate them and keep them meshed, so go ahead and use axles, bearings, and metal.

Need a Hint?

Use three axles.

15

After successfully designing a gear train with a ratio greater than 3:1, to determine the final gear ratio, count the number of rotations of the driver gear required to get one complete rotation of the driven gear.

Turn and Talk



How could you mathematically determine the gear ratio shown in Figure 7?

Develop a formula to calculate the overall gear ratio of any compound gear train that uses three axles.

NOTE: In Figure 7, gears B and C share the same axle. Engineers call these gears coaxial.

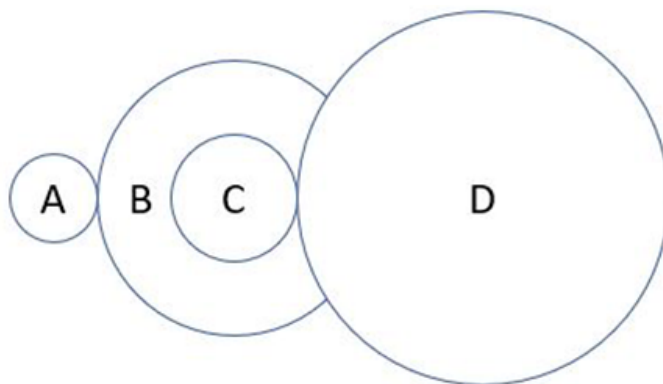


Figure 7. Representation of a compound gear train with gears labeled A (driver) through D (final

driven)

As you learned in step 8, there is a trade-off between speed and torque.

In Figure 7, the output gear rotates at $\frac{1}{9}$ the rate of the input gear.

However, the output gear rotates with 9x the torque of the input gear.

Turn and Talk



Consider Fran's tractor. Predict the relative sizes of the driver and driven gears in the tractor's transmission. (Which gear is larger?) Do you think tractors are geared to maximize speed or torque? Why would this be the case?

Go for Slow: Part 2

16

Modify your tractor to use a compound gear train. Use any combination of gears and axles available. Again, the slowest tractor wins!

Conclusion

Question 1

Is this gear train (Figure 8) designed to maximize torque or speed? How do you know?

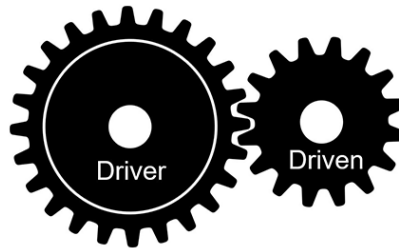


Figure 8. Example Gear Train 1

Question 2

Figure 9 shows a gear train. If gear A rotates clockwise at 700 rpm, in what direction and at what speed does gear D rotate?

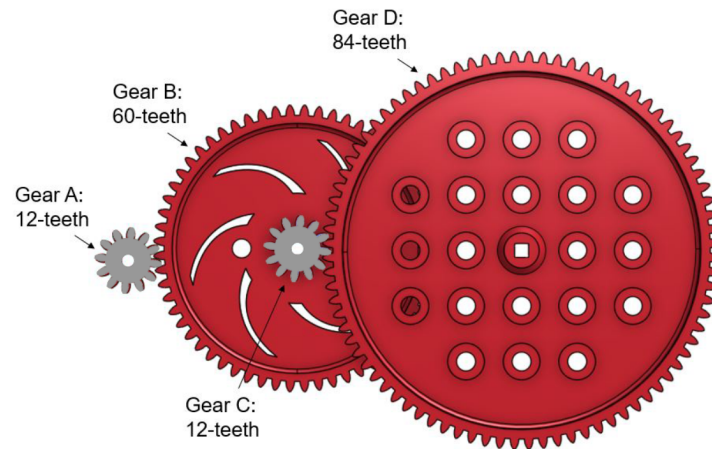


Figure 9. Example Gear Train 2

Question 3

Figure 9 shows a gear train. If 7 ft-lb of torque is applied at gear A, what is the output torque at gear D?

Question 4

Describe and compare the benefits and drawbacks of simple and compound gear trains.